



Designing membrane-targeted peptides requires understanding sequence-environment interactions. This thesis combines molecular simulation, biophysical modelling, and computational optimization to investigate these interactions. Free-energy calculations reveal how membrane composition and peptide orientation influence binding, while complementary models incorporate environmental effects into peptide conformational and sequence design within an α -helical scaffold. Classical and hybrid quantum-classical algorithms explore these formulations, linking peptide-membrane interactions with environment-conditioned sequence design. Together, these contributions provide a framework to identify peptides with specific environmental preferences and prioritize candidates for simulation and experimental validation.